

Using CAC Bar Scanner

Installing and launching from the Start menu

CAC Bar Scanner is a single self-contained program — there is no separate installer to download. Click the one-click download link on the [repo's front page](#) (the **↓ Download BarScanner.exe** button near the top of the README) or use this direct link: <https://github.com/NuclearChickens/CAC-Bar-Scanner/raw/main/BarScanner.exe>. Your browser will save it to your **Downloads** folder by default — that's exactly where you want it. Open File Explorer, click **Downloads** in the left-hand sidebar, and you'll see **BarScanner.exe** sitting there. Double-click it to run. The first time you launch, Windows may pop up a blue box that says *"Windows protected your PC"* — this is normal for any program that isn't sold through the Microsoft Store. Click the small **More info** link in that box and then the **Run anyway** button that appears; Windows will remember your choice and won't ask again. As soon as the app opens it asks whether to install for this PC — click **Install**. Windows will then pop up a blue *User Account Control* permission box, and you click **Yes**. With that one prompt the app:

- copies itself into **C:\Program Files\CAC Bar Scanner** so the Start menu has a permanent home for it,
- adds itself to the Start menu (and Settings → Apps) for every user on the computer,
- and sets up a shared folder at **C:\ProgramData\CACBarScanner** so every operator who logs in sees the same configuration and scan history.

After that you can safely delete the copy in Downloads — the Start menu and taskbar use the Program Files copy. To launch the app any time, tap the Windows key, type "Bar", and press Enter.

If you click **Not now** instead of Install, the app still runs from wherever you double-clicked it, just without a Start menu entry or the shared configuration folder. The install offer doesn't reappear for the same Windows user, but if a different user logs in and runs the exe, they'll get the same prompt. Once anyone clicks **Install**, the offer is dismissed for every user on the PC.

Uninstalling

Two ways:

- Open **Settings** → **Apps** → **Installed apps**, scroll to or search for **CAC Bar Scanner**, click the ... menu next to it, and choose **Uninstall**.
- Or open the **Start menu**, find **CAC Bar Scanner**, right-click the entry, and choose **Uninstall**.

Either way Windows runs the app one last time in uninstall mode. A small dialog appears with a single checkbox: **Also delete settings, ban list, and all logs**. Leave it unchecked to remove just the program (keeping your configuration so a future reinstall picks it back up); tick it for a full wipe. Click **Uninstall** and approve the User Account Control prompt. The Start menu entry, the Add/Remove Programs listing, and the `C:\Program Files\CAC Bar Scanner\` folder are all cleaned up — the folder finishes deleting itself on the next reboot, which is normal Windows behavior since the running uninstaller can't delete itself in real time.

The Scanner tab

This is the main screen and where the cashier or door staff lives all night. A big white banner sits at the top reading **Ready to scan**. The cursor is already parked in the input box beneath it — leave it there. When a CAC is swiped across the USB scanner, the 18-character barcode types itself into the box automatically and the verdict appears within a fraction of a second:

- **ALLOWED** (green banner). The person is cleared. The banner shows which drink number this is for them (e.g. *"3rd drink"*) along with their category and branch. The scan is counted toward their limit.
- **DENIED** (red banner). The person is not cleared. The banner shows the reason in plain English — for example *"You've had 3 drinks today (limit 3)"*, *"Bar is closed (outside operating hours)"*, *"Active Duty (AD) not allowed"*, or *"This DoD ID is banned until 2026-12-31"*. Denied scans are **not** counted, so someone can't burn through their limit by trying again.
- **INVALID SCAN** (red banner). The scanner read something that doesn't look like a CAC barcode. Usually means the card was swiped too fast or at a bad angle — just try again. Nothing is recorded.

Below the banner the **Decoded** box shows the DoD ID, category, and branch of whoever was just scanned, plus a running count of how many drinks they've had in the current session. The bottom of the screen shows the current operating window (e.g. *"Open: 17:00–02:00"*) so you can confirm at a glance that the right rules are in effect.

If the input box ever loses focus — say someone clicked away by accident — just click anywhere on the Scanner tab and it pops back. The **Clear** button at the bottom blanks out the last decoded result, which is handy between customers.

The Hours tab — when the bar counts

This tab controls *when* drinks are counted. There are two modes:

- **Operating hours** (default). You set an open time and a close time in 24-hour format. While the bar is open, every allowed scan counts toward that person's session total. When the close time passes, the bar enters *closed* state, scans are denied with *"Bar is closed"*, and counts reset

automatically at the next open. If you leave both fields at **00:00**, the app treats the bar as open 24 hours.

- **Rolling window**. Instead of bar hours, you set a window length in hours (1 to 168). Each person's count is based on however many drinks they've had in the last **N** hours — sliding continuously. Good for events that don't have fixed open/close times.

Switching between modes is a single radio button at the top of the tab. The fields that don't apply to the current mode are hidden so you don't accidentally edit them.

The Limits tab — drinks per person

One number: the maximum drinks any single person can be served in a session. Default is 3. Once a person hits that number, every subsequent scan from them flashes red with *"You've had N drinks today (limit N)"* until either the session ends (Operating hours mode), the rolling window slides past their earlier drinks (Rolling mode), or you do a manual reset on the Reset tab.

The Roster tab — who's eligible

Two side-by-side checklists:

- **Categories (PCC)** — Active Duty, Reserve, Retiree, Dependent, Contractor, etc.
- **Branches** — Army, Navy, Marines, Air Force, Space Force, Coast Guard, and the various civilian/contractor codes.

A scan is allowed only if **both** its category and branch are checked here. **All** and **None** buttons at the top of each column flip every box at once, which is handy when you want to start from "deny everything" and tick on just the few you want, or vice versa.

The Banned tab — block specific people

Sometimes you need to deny one specific person regardless of the roster. Type their 10-digit DoD ID into the **EDIPI** box. Leave the **Expires** box blank for a permanent ban, or type 8 digits as **YYYYMMDD** (e.g. **20261231** for "until 31 December 2026"). Click **Add**. Bans show up in the list above; click one and hit **Remove selected** to lift it. Expired bans show **(EXPIRED)** next to them in the list but stop denying scans automatically on their expiry date — you don't have to remember to remove them.

The Reset tab — wipe today's counts

This is for moments like a shift change or a manual "start the night over". Click **Reset drinks for the day**; the tab asks for two different CAC scans to confirm. After both authorizers swipe, every existing drink count is cleared and the reset shows up in the **Public reset log** below — that log is permanent

and lists who authorized each reset along with the date and time, so there's always an accountability trail.

The Logs tab — what happened and when

A read-only feed of every administrative event: unlocks, lock-outs, settings changes, ban additions/removals, manual resets, and backup exports. You cannot edit or delete entries. Hit **Refresh** to reload after something just happened. This is the place to look when someone asks “who changed the limit last Thursday?”.

The Backup tab — saving and restoring

The app keeps all of its settings, the ban list, and every log under `C:\ProgramData\CACBarScanner\` on Windows (`~/ .cac_scanner/` on Linux/macOS). This is a **shared** folder — every Windows user on the kiosk PC sees the same hours, limits, roster, bans, and scan history, so any operator who launches the app picks up exactly the same configuration. Updating BarScanner does not touch that folder, so your setup survives upgrades. The Backup tab is there for two cases:

- **Export full backup** packs everything into a single `.zip` file. Stash it somewhere safe (the file contains DoD IDs and the ban list — treat it like an HR document).
- **Import full backup** is for moving your setup to another machine, or for restoring after a disk failure. It **completely replaces** the current settings and logs on the machine — there's a confirmation dialog before anything is overwritten. Import is hidden behind the same 2-CAC unlock as other admin actions.

Unlocking settings (the lock banner)

For everyday scanning you don't need to do anything special — the Scanner tab is always available. Anything that changes configuration (Hours, Limits, Roster, Banned, Backup → Import) sits behind a lock. Each of those tabs shows a pink banner at the top reading **LOCKED — scan 2 CACs to enable editing**. Click **Unlock**, a small window pops up, and two different people swipe their CACs through the scanner. The banner turns green showing which two IDs authorized the session. From that point you have **5 minutes** to make changes before the app auto-locks again; you can also press the **Lock** button to lock it back immediately. Everything you change while unlocked is recorded in the Logs tab against both authorizers.

Handy keyboard shortcuts

- **F11** — toggle fullscreen (kiosk mode)
- **Esc** — exit fullscreen

- **Ctrl+Q** — quit the app
- **Enter** — finish a scan manually (most scanners do this for you)

Where the app stores its data

Everything lives in **C:\ProgramData\CACBarScanner** — Windows's standard location for machine-wide application data. The folder is created during the one-time admin install and granted read/write access for every user on the PC, which is what makes the configuration shared:

- **settings.json** — your configuration
- **scans.jsonl** — running drink counts
- **audit.jsonl** — settings changes
- **resets.jsonl** — manual resets

You don't normally need to touch any of these — the app reads and writes them automatically and the Backup tab is the supported way to copy or restore them.